

Neutron diffraction study of sodium borosilicate waste glasses containing uranium

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Abstract

The effect of uranium oxide on the structure of sodium borosilicate host glasses has been studied by neutron diffraction. The samples were prepared by quenching the melted mixtures of composition 70 wt% $[(65 - x)\text{SiO}_2 \cdot x\text{B}_2\text{O}_3 \cdot 25\text{Na}_2\text{O} \cdot 5\text{BaO} \cdot 5\text{ZrO}_2] + 30 \text{ wt}\% \text{UO}_3$ with $x = 5, 10$ and 15 mol%. It was found, that the U-loaded glasses possess good glass and hydrolytic stability. An enhanced probability for inter-mediate atomic correlations at around 4.8 Å has been established. The RMC simulation of the neutron diffraction data is consistent with a model where the uranium ions are incorporated into interstitial voids in the essentially unmodified network structure of the starting host glass. The U–O atomic pair correlation functions show a sharp peak at around 1.7 Å, and several farther distinct peaks are at 2.8, 3.6 and 4.1 Å. The uranium ions are coordinated by six oxygen atoms in the 1.6–3.4 Å interval.

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1. Introduction

Alkali borosilicate glasses are of significant current interest as suitable materials for isolating host media for radioactive waste material storage (i.e. UO_3 or PuO_2) [1]. Structural characterization of these glasses is essential for understanding of glass durability. We are motivated in the investigation of multi-component sodium borosilicate waste (host) glasses with the general composition of $(65 - x)\text{SiO}_2 \cdot x\text{B}_2\text{O}_3 \cdot 25\text{Na}_2\text{O} \cdot 5\text{BaO} \cdot 5\text{ZrO}_2$, $x = 5$ –15 mol% added with UO_3 or CeO_2 (Ce is considered as non-radioactive surrogate for Pu), with the aim to clear up the correlation between structural characteristics and their

thermal and glass stability. In host glasses SiO_2 and B_2O_3 are strong network formers; Na_2O serves as network modifier; while BaO serves both as network modifier, glass and hydrolytic stabilizers. In the course of our previous work [2] we have established that addition of ZrO_2 improves the glass and hydrolytic stability due to its strong charge compensating ability. Our results on the network structure of the host glasses obtained from high momentum transfer neutron diffraction experiment up to 30 Å^{-1} are presented in [3].

Here we investigate the glass forming ability of the host glass loaded with uranium oxide, and its effect on the structure. Neutron diffraction measurements up to 10 Å^{-1} are presented on a newly synthesized multi-component sodium borosilicate glassy system of the above composition added with UO_3 . Reverse Monte Carlo simulation is applied to characterize the atomic pair correlations, with special respect on the U–O neighbor distribution. Details of glass

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preparation procedure and prompt gamma activation analysis applied for elemental composition determination is also described.

2. Experimental

2.1. Sample preparation

The glassy samples of composition 70 wt% $[(65 - x)\text{SiO}_2 \cdot x\text{B}_2\text{O}_3 \cdot 25\text{Na}_2\text{O} \cdot 5\text{BaO} \cdot 5\text{ZrO}_2] + 30 \text{ wt}\% \text{ UO}_3$ with $x = 5, 10$ and $15 \text{ mol}\%$ (hereafter referred to as UB5, UB10 and UB15) were prepared by melt quench technique. The raw materials used were all of p.a. grade: SiO_2 , Na_2CO_3 , UO_3 (supplied by Reactivul, Bucuresti), BaO and ZrO_2 by Merck (Darmstadt), B_2O_3 by Sigma–Aldrich Co. (Hungary). B_2O_3 was isotopically enriched in ^{11}B in order to reduce the influence of the high neutron absorption of ^{10}B present in natural boron. The ^{11}B isotope enrichment was 99.6% as determined by Inductively Coupled Plasma Mass Spectroscopy (ICP-MS) technique [4].

The UB5, UB10 and UB15 glasses were synthesized by melting the previously homogenized powder mixtures in a high temperature electrical furnace with a platinum crucible under atmospheric conditions at 1500, 1450 and 1400 °C, respectively. The melted mixture has been kept at the melting temperature for 2 h, meanwhile the melt was periodically homogenized by mechanical stirring. Thereafter the melt was cooled to the pouring temperature of 1450, 1350 and 1300 °C for UB5, UB10 and UB15 compositions, respectively, and kept there for 30 min. Finally, it was quenched by pouring the melt on a stainless steel plate. The specimens were kept in exsiccator to avoid hydrogen absorption.

Powder samples were prepared by powder milling of the quenched glasses in an agate mill.

2.2. Prompt Gamma Activation Analysis measurement

The elemental composition of the specimens was verified by Prompt Gamma Activation Analysis (PGAA) spectroscopy using cold neutrons at the Budapest research reactor. This nuclear analytical technique is based on the detection of prompt gamma radiation emitted by the sample while being irradiated in a neutron beam [5]. The elemental compositions were determined based on the data library given in [5]. The spectra were evaluated using the code Hypermet-PC experiments [6]. The mass ratios of the components were calculated from the peak area ratios corrected by the counting efficiencies. The concentrations given in Table 1 were determined based on the assumption that all significant components appear in the spectra. The oxygen content, however, was calculated from stoichiometry, as the analytical sensitivity is rather low for this element. The sum of the masses for the identified oxides was normalized to 100%. The specimens did not contain any hydrogen.

Table 1

Elemental composition (at%) of the uranium containing sodium borosilicate glasses measured by PGAA method

Element	Elemental composition (at%)		
	UB5	UB10	UB15
Si	15.27 (2)	14.4 (2)	15.99 (2)
B	3.43 (3)	5.97 (2)	9.10 (1)
Na	12.76 (2)	12.7 (2)	11.8 (2)
O	61.05 (0.5)	59.84 (0.5)	55.5 (0.4)
Ba	1.34 (3)	1.40 (4)	1.23 (4)
Zr	2.81 (5)	2.45 (7)	2.96 (6)
U	3.11 (2)	3.14 (4)	3.3 (3)

The relative errors are indicated in brackets.

2.3. Neutron Diffraction experiments

Neutron Diffraction (ND) measurements have been performed at the 10 MW Budapest research reactor using the ‘PSD’ neutron powder diffractometer [7]. Monochromatic wavelength of $\lambda_0 = 1.068 \text{ \AA}$ was used. The diffraction spectrum was measured in the momentum transfer range of $Q = 0.95\text{--}9.8 \text{ \AA}^{-1}$. The powder specimens of about 3–4 g were filled in cylindrical vanadium sample holder of 8 mm diameter, 50 mm height and 0.07 mm wall thickness. The specimens had to be handled with a special care due to their radioactivity. Correction and normalization procedures utilized to obtain the total structure factor $S(Q)$ from the measured pattern was described in our previous work [8]. Fig. 1 shows the experimental $S(Q)$ for the three compositions (the results of RMC modeling is also indicated, the details will be discussed in the next section).

The ND pattern show, that the specimens are fully amorphous, and no hydrogen was detected neither by ND nor by PGAA measurements, meaning that the applied six-component matrix glass is an effective host for embedding the large uranium ions and they are hydrolytically stable.

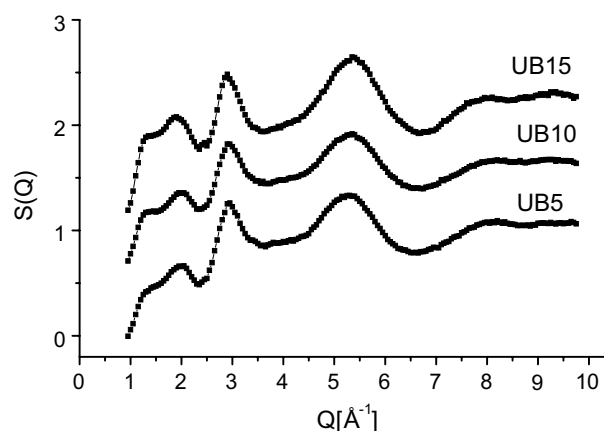


Fig. 1. Neutron diffraction structure factors of uranium containing sodium borosilicate glasses: experimental data (squares) and RMC simulation (solid line). (The curves are shifted vertically for clarity.)

Table 2
Several weighting factors (%) for the glassy samples

Atom pairs	Weighting factor (%)		
	UB5	UB10	UB15
Si–O	15.41	14.03	14.67
B–O	5.55	9.33	13.03
O–O	43.10	40.78	34.64
Si–Si	1.37	1.20	1.46
Na–O	11.28	10.81	9.21
Ba–O	1.64	1.66	1.32
Zr–O	4.90	4.13	4.55
Si–Na	2.01	1.86	1.89
Si–B	0.99	1.60	2.68
U–O	6.30	6.20	6.00

In dependence of boron content slight changes may be observed in the $S(Q)$'s. Especially, the small (pre)peak at $1.3 \text{ \AA}^{-1}$ gets more pronounced with increasing boron composition. The positions of the next peaks are at around 1.9, 2.9, 5.3 and $8.0 \text{ \AA}^{-1}$. The total structure factor is the weighted sum of the partial structure factors [2]. Table 2 collects those atom pairs for which the weighting factor is above 1%.

3. Reverse Monte Carlo modeling

The ND experimental $S(Q)$ data have been simulated by the RMC method [9]. For the RMC starting model a disordered atomic configuration was built up with a simulation box containing 5000 atoms, and box length of $20 \text{ \AA}$. The initial configuration was prepared from a completely random distribution of atoms in two main steps. As a first step of simulation procedure, the MCGR method was applied [10]. MCGR is a one-dimensional version of RMC to produce a total atomic pair correlation function, $g(r)$, allowing atomic movements to minimize the difference between model and experimental structure factors. The aim of MCGR simulation is to lighten the atomic motions in RMC calculations. We have used a cut-off constraint of $1.1 \text{ \AA}$ for each atomic pair in the MCGR simulation.

Fig. 2 shows the total atomic pair correlation functions, $g_{\text{MCGR}}(r)$ obtained from MCGR simulation. Note, that due to the limited Q -range of the present experiment the r -space resolution is rather low, $\Delta r = \frac{2\pi}{Q_{\text{max}}} \approx 0.6$, and therefore the Si–O and B–O first neighbor atomic pair distributions are not resolved in the first peak of $g_{\text{MCGR}}(r)$ in contrast to the high Q -range ND experiment on host glasses reported in Ref. [3].

Several density values were tested between 0.07 – $0.085 \text{ at \AA}^{-3}$. The best fit of the experimental ND pattern was revealed with $\rho_0 = 0.078 \pm 0.002 \text{ at \AA}^{-3}$, which is about 7% higher than $0.073 \text{ at \AA}^{-3}$ obtained for the host glasses [3].

In RMC modeling several constraints are used to obtain reliable three-dimension atomic configuration, described by the partial atomic pair correlation functions, $g_{ij}(r)$. The number of $g_{ij}(r)$ is 28 of the present seven-component

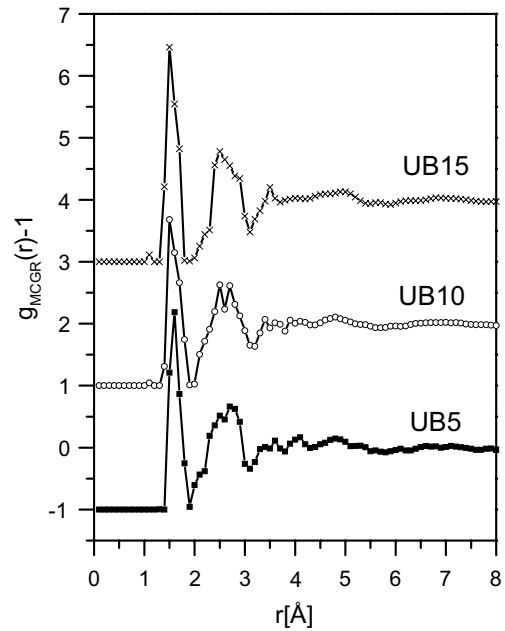


Fig. 2. Total atomic pair correlation function of uranium containing multi-component glasses obtained from MCGR simulation of ND data presented in Fig. 1. (The curves are shifted vertically for clarity.)

glasses; their determination from the present ND experiment is fairly impossible. However, based on a priori structural considerations we may expect useful information on the structure. As far as, the $S(Q)$'s of the UO_3 -loaded specimens and the corresponding host glasses [3] look fairly similar, as it is illustrated in Fig. 3, it is reasonable to suppose that uranium does not change significantly the basic network former units of the host glass. Therefore, for the present RMC modeling we have used the same constraints as for the host glasses [3]. The following distances of closest approach and connectivity constraints have been applied – B–O: 0.8 – 1.9 (first neighbor interval), Si–O: 1.5 – 1.9 (first

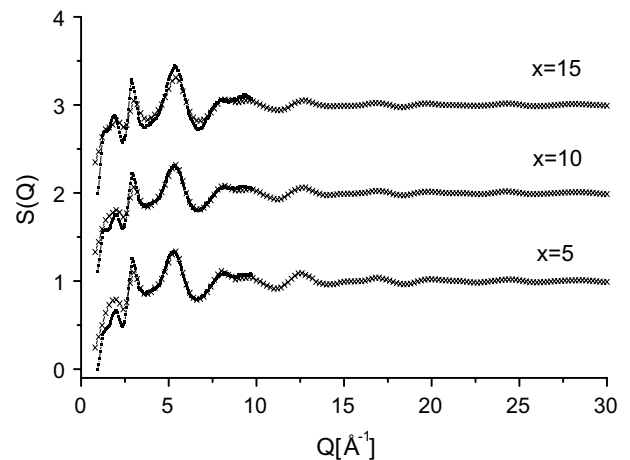


Fig. 3. Structure factor of uranium containing glasses measured up to $9.8 \text{ \AA}^{-1}$ (squares) and that of the corresponding host glasses up to $30 \text{ \AA}^{-1}$ (crosses) [3] in dependence of boron content. (The curves are shifted vertically for clarity.)

neighbor interval), Zr–O: 1.9, Na–O: 2.05, O–O: 2.15, Si–Na: 2.4, Si–B: 2.45, Ba–O: 2.45, Si–Si: 2.8, Na–Na: 3.0 Å. Si atoms were forced to have four oxygen neighbors, while for B atoms we have allowed both three and four-coordinated surroundings. For the first neighbor cut-off distance of U–O atom pairs we have applied 1.6 Å, not to overlap with Si–O distribution, but to allow 1.8 Å as a short U–O distance reported for uranyl $[\text{UO}_2]^{2+}$ ions embedded in glasses (e.g. [11–13]). In this study we did not use any further constraints, although, it is widely accepted that two types of oxygen atoms are present in alkali silicates (e.g. [14,15]); the bridging (BO) and non-bridging (NBO) oxygen atoms, and both for the Si atoms and for the cations the distance to BO or NBO is reported to be different. In our case this may lead to different U–O distances as well, although the mechanism is different. As a result of RMC simulation the obtained partial distribution functions do not reflect these differences without a priori constraints.

4. Results and discussion

Fig. 3 compares the structure factors for the uranium containing samples with those of the corresponding host glasses [3]. The overall character of the glassy structure seems to be fairly similar, although slight differences may be seen, especially in the relatively low Q -range. The broad first asymmetric peak of the host glasses at around 2.0 Å^{-1} splits into two characteristic peaks centred at 1.3 Å^{-1} and 1.9 Å^{-1} in the case of the U-containing specimens. With increasing boron content the intensity of the peak at 1.3 Å^{-1} becomes more pronounced, indicating an enhanced probability for correlation of atomic arrangements at distances around $2\pi/Q$ ($=1.3 \text{ Å}^{-1}$) $\approx 4.8 \text{ Å}$. This establishment is consistent with the peak at around 4.8 Å in $g_{\text{MCCR}}(r)$ as shown in Fig. 2.

The final RMC fit matched reasonable well the experimental data as it is illustrated in Fig. 1. The rather low r -space resolution of the present experiment, however, did lead to unresolved peak distribution in several $g_{ij}(r)$, i.e. for B–O first neighbors for which we have found well resolved first neighbor distances at 1.40 and 1.60 Å for the corresponding host glasses from high Q -range ND experiments [3]. Therefore, here we focus our interest on the U–O atomic pair correlation function, while for the other $g_{ij}(r)$'s we refer to Ref. [3]. The weight of the U–O atom pairs in the ND experiment is around 6% (see Table 2) thus we may expect to obtain reliable results for $g_{\text{U-O}}(r)$. Fig. 4 displays $g_{\text{U-O}}(r)$ for the three specimens. Obviously, they are very similar to each other. A sharp peak appears at around $1.70 \pm 0.05 \text{ Å}$, and for higher distances several weak peaks may be observed at about 2.8, 3.6 and 4.1 Å (see Fig. 4 inset). The overall run of $g_{\text{U-O}}(r)$ proved to be stable. The first sharp peak has to be handled carefully because it partially overlaps with the Si–O peak (centred at 1.60 Å), and with B–O distribution. In order to avoid the possible errors originating from the overlapping

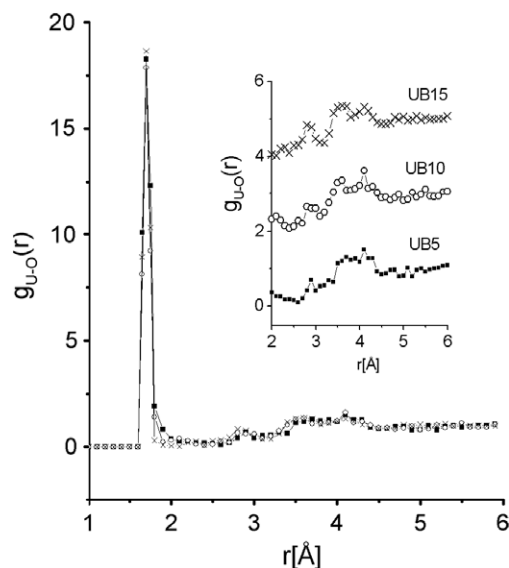


Fig. 4. U–O partial correlation function for the uranium containing sodium borosilicate glasses obtained from RMC simulation: UB5 (square), UB10 (open circle) and UB15 (cross). The inset shows the small intensity peaks above from 2 Å on an enlarged scale.

distributions, we have calculated the average coordination numbers for the corresponding partial atomic correlation functions from radial distribution function analyses. It was revealed that the average coordination number for Si–O is 3.94, and for B–O 3.5, 3.1 and 3.1 for the UB5, UB10 and UB15 specimens, respectively. These values are very close to the coordination number values for the corresponding host glasses [3], thus we may conclude that the first sharp peak of $g_{\text{U-O}}(r)$ corresponds to U–O first neighbor distance, and its artificial character may be excluded. This suggests that uranium ions are surrounded in a relatively short distance by 2–3 oxygen atoms, and further oxygen atoms are bonded at higher distances. This observation highly supports the tendency of forming uranyl ions reported in the literature [11–13].

Several coordination number distributions, $\text{CN}_{ij}(n)$ were analysed from RMC modeling. Fig. 5 displays the Si–O, B–O, Na–O and U–O coordination number distributions. The displayed distributions (and also the number of average coordination number) depend on the range of analysed distance; therefore, here we give the corresponding intervals as well. We have revealed the following average values with an error of about 1% for Si–O, and 3% for B–O, Na–O and U–O coordination numbers: $\text{CN}_{\text{Si-O}} = 3.94$ calculated between 1.4 and 2.0 Å, $\text{CN}_{\text{B-O}} = 3.5, 3.2$ and 3.1 for UB5, UB10 and UB15 specimens, respectively, calculated between 1.4 and 2.0 Å, $\text{CN}_{\text{Na-O}} = 5.8$ calculated between 2.05 and 2.8 Å and $\text{CN}_{\text{U-O}} = 6$ calculated in the 1.6–3.4 Å interval. For the network former Si–O and B–O we have revealed similar data as for the corresponding host glasses [3] in accordance with the starting model. Si atoms are coordinated by nearly four oxygen atoms, while boron atoms are 3- and 4-fold coordinated. With increas-

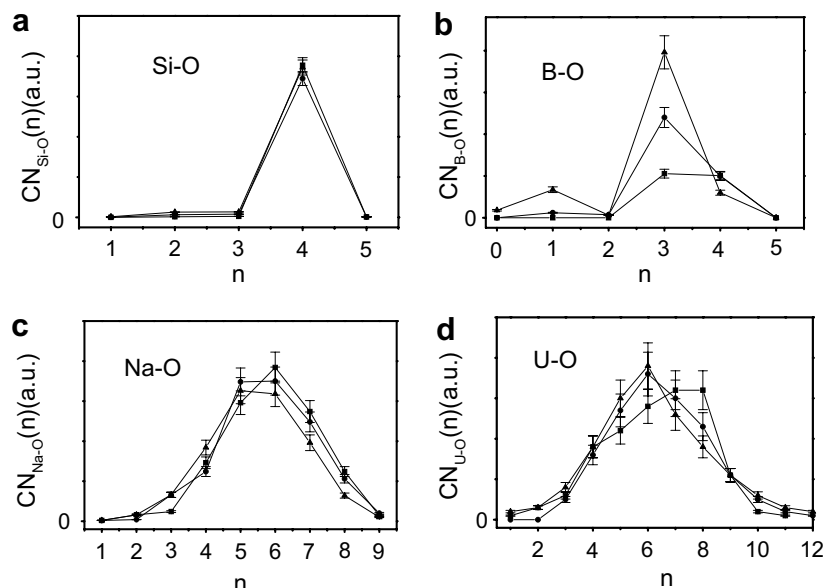


Fig. 5. Several coordination number distributions for the three U-containing glassy specimens obtained from RMC modeling, where UB5 (square), UB10 (circle) and UB15 (triangle). The error bars correspond to Si–O (5%), B–O (10%), Na–O (10%) and U–O (15%).

ing boron content the number of three-coordinated boron atoms increases, while the four-coordinated B–O surrounding is nearly unchanged. The Na–O coordination number distribution for the three samples is practically the same, and the distribution peak is centred at around six, while the U–O distribution shows for the UB15 sample a somewhat shifted distribution to higher values, but taking into consideration the limited accuracy of this experiment we cannot state that this is a real effect.

5. Conclusion

The effect of uranium oxide on the structure of sodium borosilicate host glass has been studied by neutron diffraction. Our main findings are as follows:

- the glasses possess good glass and hydrolytic stability;
- an enhanced inter-mediate range order has been revealed indicating the probability for correlation of atomic arrangements at distances around 4.8 Å;
- the RMC simulation of the ND data is consistent with a model, where the uranium ions are incorporated into interstitial voids in the essentially unmodified network structure of the starting host glass;
- the U–O atomic pair correlation functions show a sharp peak at a relatively short distance ~ 1.7 Å, and several farther smaller intensity but distinct peaks are at 2.8, 3.6 and 4.1 Å;
- the uranium ions are coordinated by six oxygen atoms in the 1.6–3.4 Å interval.

In order to clear up in more details the short-range structure, and to obtain more accurate results, the exten-

sion of the momentum transfer range of the present ND experiment is essential.

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